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Techno-economic study of the impact of blackouts on the viability of connecting an off-grid PV-diesel hybrid system in Tanzania to the national power grid

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ABSTRACT

National electrification plans for many countries with a low level of electrification promote the implementation of centralized and decentralized electrification in parallel. This paper explores different ways of utilizing an established off-grid PV-diesel hybrid system when the national grid becomes available. This is a rather unique starting point within the otherwise well-explored area of rural electrification. With particular focus on the impact of blackouts in the national grid, we evaluate the economic viability of some alternatives: to continue to use the off-grid micro-grid, to connect the existing micro-grid with or without battery backup to the national grid, or to use the national grid only. Our simulation results in HOMER demonstrate that with a grid without blackouts, there are few benefits to maintain the existing system. Low grid-connection fees, low tariffs and low revenues from selling excess electricity to the grid contribute to this result despite the fact that the system does not carry any investment costs. With a grid with blackouts, it is beneficial to maintain the system. The extent of blackouts and the load on the system determine which system configuration is most feasible. The results make clear the importance of taking blackouts in the national grid into consideration when possible system configurations are being evaluated. This is rarely quantified in studies comparing different electrification alternatives, but deserves more attention.

1. Introduction

Since the year 2000, the economy in sub-Saharan Africa has more than doubled [1,2]. Two thirds of the 950 million people living in sub-Saharan Africa lack access to electricity, and millions are connected to unreliable grids [3,4]. According to the International Energy Agency (IEA), "Efforts to promote electrification are gaining momentum, but are outpaced by population growth" [2]. Closing the electricity gap is a complex and multidimensional challenge that includes supply-and-demand related challenges in grid-connected areas as well as lack of access in off-grid areas [4].

Most governments in sub-Saharan Africa have developed national electrification strategies [1]. These mostly contain two tracks to be implemented in parallel. On the centralized track, electrification occurs primarily by extension of the national grid and is undertaken by governmental entities. The decentralized track promotes distributed solutions such as micro- and mini-grids carried out by non-governmental entities. Most strategies, however, lack details on how the decentralized track and the interplay between the two tracks are to be implemented. Without recommendations and policies for this interplay, there is a risk of decentralized systems being abandoned when the national grid becomes available, leading to a reluctance to invest in these [5]. To address some of the challenges, and upon request from the Africa Electrification Initiative, The World Bank has created a comprehensive guide focusing on regulations and policies needed in order to create commercially viable small power producers and mini-grids in rural areas, of typically 100 kW or more [1]. The guide also discusses technical and economic frameworks for the interconnection of such systems with the national grid.

The vision of the Tanzanian government is to move Tanzania from being a low- to a middle-income economy by 2025 [6]. Electricity is recognized as important for that development. Access to electricity has increased from 13% in 2008 to 36% in 2015, yet with significant differences between urban and rural areas [2,7,8]. According to the Power System Master Plan [7], Tanzania shall reach 60% electrification by 2020, and 90% by 2035. Important factors for the increase have been an extensive expansion of the national grid in combination with a substantial reduction in connection fees [2]. Initiatives promoting decentralized solutions include the UN initiative Sustainable Energy for All [9] and Power Africa initiated by US AID [10]. The Small Power

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Producer Framework [11] is encouraging private-sector investments in renewable energy, and the Rural Energy Agency [12] focus on modern energy services in rural areas.

According to The World Bank statistics [13,14], customers on the Tanzanian national grid experience on average nine occasions of blackouts per month. In 2013, the duration of a typical blackout was 6.3 h resulting in a total of 54 blackout hours per month, declining from 71 h in 2006 [14,15]. The length and frequency of the blackouts, however, vary depending on the time of year and location.

To address the issue of electricity access, numerous studies on decentralized electrification have been conducted. Commonly, solutions include micro- and mini-grids supplied by PV or diesel generators, or hybrid systems combining the two. Mandelli et al [16] and Chaurey and Kandpal [17] provide comprehensive reviews of scientific work addressing decentralized rural electrification. Case-specific design studies (see [18–23] for recent examples) and reviews of methodologies and software tools for design and optimization of single-source and hybrid systems [24–27] are common. Operation and control optimization of hybrid systems is another studied field [28–30]. Goel and Sharma [31] present a review of research on performance analysis of hybrid systems. Bahamara et al [32] provide a review of research contributions using the software tool HOMER [33] for planning hybrid renewable energy systems. Another common field of investigation is whether to supply electricity to a location by means of centralized or decentralized solutions [18,34–36]. Some researchers use the economic distance limit as a tool for choosing between decentralized electrification and grid extension [18,37,38].

Although they commonly occur in regions with a low level of electrification, blackouts are rarely considered in studies comparing centralized and decentralized electrification or in studies of grid-connected micro- and mini-grids. Cader [35] assesses the quality of grid supply in terms of blackouts, transmission and distribution losses, and connection charges in her analysis of challenges associated with centralized electrification. Her conclusion is that these challenges require more attention. Mahapatra and Dasappa [38] consider off-grid PV, biomass gasifier-based power generation and grid extension in the investigation of electrification strategies for village electrification in India. They consider three levels of electricity availability (6, 8 and 12 h/day), and show how the economic distance limit for different off-grid system solutions depends on load level and electricity availability. They find power generation through biomass gasification more cost competitive than PV.

Twaha et al [39] have explored the possibilities of using PV to supplement the national grid in Uganda. Currently, diesel power plants are used to reduce the gap between supply and demand of electricity, resulting in high electricity prices and a high environmental impact. The authors argue that if incorporated into the national electricity plan, financial support towards PV can be expected. Using HOMER, the study explores the economic effects at the village level of installing different combinations of PV, batteries and diesel generator to be used together with existing grid connection. Despite the fact that using only the grid has been found to be the most economically viable alternative for the village, the authors conclude that with expected economic incentives, grid-connected PV can become economically viable at village level in the future and at the same time contribute to decreasing energy prices and have an environmental impact at a national level. In [40], Murphy et al study the effect of blackouts in the national grid on the work presented in [39]. Based on figures for average blackout frequency and duration in Uganda given by the World Bank, they create a pseudo-random pattern for blackouts. They re-evaluate optimized system configurations in [39], and still find that the grid is the most cost-competitive solution, yet it cannot deliver uninterrupted electricity. Complemented with PV and batteries, the electricity availability in the village increases from 82.3 to 90.5% for an additional 0.07 USD/kW h [40]. By optimizing system configurations to meet different availability constraints, the authors further explore the cost of increased availability. Their results indicate that with current prices, a diesel generator is the most economical choice for a backup system for the Ugandan village. Expected increase in fuel price and decrease in PV-system component costs, however, lead the authors to conclude that PV-diesel hybrid systems will be economically competitive in the long run.

Rajbongshi et al [37] evaluate different system alternatives for electrification of an Indian village. They consider grid-connected and off-grid hybrid systems using PV, biomass gasifier (BG) and/or diesel and conventional grid extension. They use HOMER to optimize the system for different load profiles and consider four different levels of grid availability (6, 12, 18 and 24 h/day). They find that an optimized BG-grid system can save roughly 50% of the cost of electricity relative to an optimized PV-BG off-grid system. The cost of energy (COE) for the grid-connected configurations was nearly the same for all considered variations (0.064–0.067 USD/kW h). In some cases, the COE decreased upon decreasing grid availability, a result linked to the possibilities of selling excess power at times of overproduction. Important to notice is that the optimization process here results in different system configurations for different load and grid availability levels. The results therefore show the importance of considering grid availability in order to obtain optimization results which correspond with the actual conditions.

Khoury et al [41] present the design of a backup PV-battery system intended to supply electricity to a single-family house in Lebanon during two different levels of daily blackouts. They use Genetic Algorithm and Particle Swarm Optimization programmed in MATLAB [42] for comparison and validation of different system configurations. The optimal PV and battery sizing is determined based on the lowest overall cost. In [43,44], the same authors present load-management systems that are applied during blackouts for predictable and unpredictable loads respectively, implemented in a residential house using a PV-battery backup system to replace the grid during blackouts. They conclude that by applying a load-management system to a PV-battery backup installation, system operation can be optimized and great sizing reductions can be achieved, resulting in considerable economic savings.

Despite numerous studies and publications addressing various aspects of rural electrification, little attention has been given to how to deal with existing off-grid power systems upon the extension of the national grid. Blackouts in the national grid have been studied merely with focus on suitable backup and grid-supporting solutions, while the role of existing systems has not been examined. This study aims to fill a research gap by providing a case study of a PV-diesel micro-grid (PVDMG) being retro-fitted to a national grid with frequent blackouts. Using HOMER [45], we examine the economic viability of different alternatives that the owner of a particular PVDMG already in operation in Tanzania has when the national grid enters an area: continuous operation of the off-grid PVDMG, grid only, or grid connection of the PVDMG with or without batteries. We study the effect of different parameters when investigating the interplay between centralized and decentralized power systems, and point at aspects that deserve attention in studies of similar cases.

The remainder of this paper is organized as follows: it firstly gives a systematic introduction to the research methodology in Section 2, including information about the studied system, the way it has been modelled and the simulated variations. Section 3 presents the results of the simulations. In Section 4, limitations and sensitivity of results are discussed. Section 5 contains our conclusions.

2. Research methodology

Fig. 1 provides an overview of the used research methodology. To find an answer to the research question *How can the existing PVDMG be utilized together with grid connection?* A model of the existing PVDMG has been created using the software HOMER – a powerful standard tool for the economic analysis of micro-grids [45]. We have varied boundary

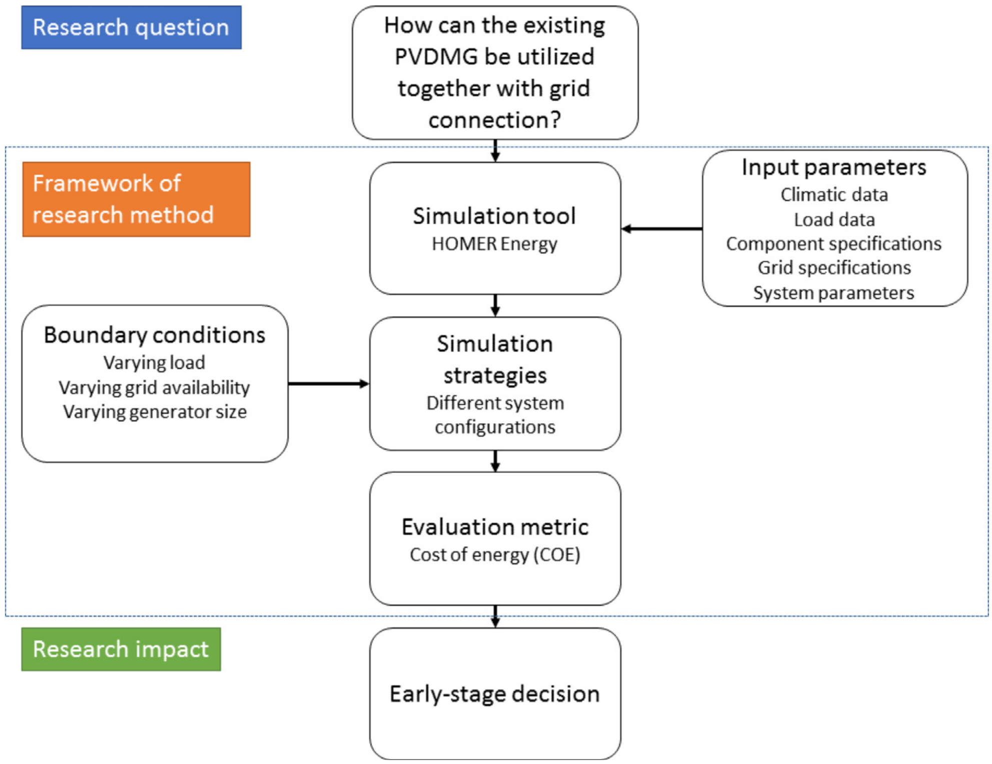

Fig. 1. Overview of the used research methodology.

conditions and applied different system configurations in order to investigate the effect of load, generator size and blackouts in the national grid on the feasibility of different possible modifications of the PVDMG. The system configurations primarily include to keep various parts of the PVDMG, and sub-variants include the selling of excess electricity to the grid. The evaluation is based mainly on cost of energy (COE), and the results show the viability of different possible system configurations.

We have chosen to model a real, existing PVDMG. By not being optimized in terms of component selection, sizing and system control, our study carries a rather important message by providing a realistic picture of how a real system that would be considered for connection to a national grid may look. Despite the case-specific nature of the figures and numbers presented as results, the overall trends and aspects to study and consider for similar cases are hence highly relevant.

2.1. System description

2.1.1. The modelled case

The studied PVDMG is owned by a Community Based Organization and is used at a community centre near Mwanza in Tanzania. At the centre, the organization runs computer courses and vocational training; it has a pre-school; it hosts meetings; and it offers typing and printing services. The PVDMG was built in 2008, in conjunction with the establishment of a carpentry workshop and the purchase of a generator to power workshop machines. An existing 12 V PV system was complemented with additional modules, and the generator was used to form a hybrid system. A 12 V DC distribution system supplies lights, and 230 V AC distribution is used to power a meeting hall with a TV, a small shop, a computer room and classrooms. Fig. 2 shows a schematic drawing of the PVDMG.

Recently, the national power grid was extended to the location. The organization is interested in limiting the generator use by connecting the PVDMG to the national grid. They wish to keep the PVDMG for reliability purposes, and believe it is economically feasible to continue using the already installed system instead of purchasing all needed electricity from the grid. In 2011, Dalarna University installed a system for monitoring the PVDMG for research purposes [46], designed according to the IEC standard 61,724 [47]. It includes the measurements also shown in Fig. 2. The monitored data has been used to determine load profiles and component- and system-efficiency factors used in the model.

2.1.2. System components

The components and loads forming the studied PVDMG are specified in Table 1. The generator primarily serves a carpentry workshop, and supplies power to the hybrid system via a manual switch.

2.2. Simulation tool – HOMER

For modelling, we have used HOMER [45]: a powerful simulation tool developed especially for the techno-economic analysis of microgrids and hybrid systems. HOMER has over 100 000 users worldwide, and is widely used for scientific studies [33]. It has some limitations in terms of detailed technical modelling. As shown in [48], proper technical modelling of the studied PVDMG is possible when system and component parameters can be derived from measurements. The main intention with HOMER is to optimize on- and off-grid hybrid systems based on economic viability and power availability. The user provides the model with parameters describing considered system configurations, component specifications, costs, resource availability, energy demand and operational boundary conditions. System operation is simulated by energy balance calculations for each hour of one year. After simulating possible system configurations, HOMER generates a list of design options with the lowest net present cost (NPC) and COE for each system configuration.

2.3. Evaluation criteria

We use COE as the main parameter for the evaluation of simulation results. COE represents the unit energy costs of establishing and operating an energy system over its entire life. The parameter allows for straightforward comparisons of costs for different system variations.

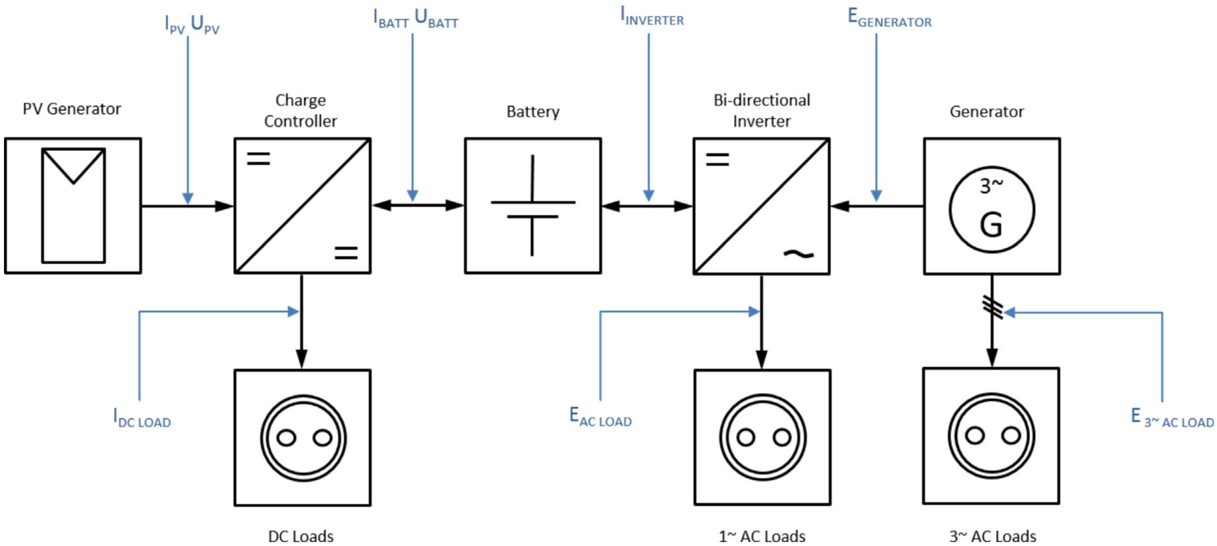

Fig. 2. A schematic drawing of the PVDMG and the measurements in the monitoring system.

COE is defined in HOMER as:

$$COE = \frac{C_{ann,tot}}{E_{served}} \tag{1}$$

where $C_{ann,tot}$ is the annualized value of the total NPC and E_{served} is the total amount of energy that went towards serving the loads during the year, plus energy sold to the grid. The NPC and hence the COE includes capital costs, replacement costs, operation and maintenance costs, fuel costs, capital cost, grid power purchase, and revenues including salvage value and grid sales. HOMER allows for sensitivity studies for various parameters such as component and fuel/grid electricity costs, but capital costs are fixed for the whole project lifetime. Continuous degradation of system performance cannot be modeled in HOMER, but has been taken into consideration in our model by specifications at the component level. We present our results as savings in COE for each evaluated system configuration relative to continuous operation of the off-grid PVDMG without modifications (base case) as:

$$S_{COE} = COE_{system} - COE_{basecase} \tag{2}$$

2.4. Input parameters for modelling

The technical and economic parameters presented in Table 2 have been used as input parameters to HOMER, together with the component specifications in Table 1. Further explanations of the parameters can be found in this chapter, together with information on load profiles.

2.4.1. Modelling of components

As the PVDMG has been in operation for some years, the assumed project time was shortened from the commonly used 25 or 30 years to 20 years. For already available components, no investment cost was applied. For components requiring replacement, prices obtained from local retailers at the last replacement (2011–2015) were assumed. Since then, equipment costs (PV, inverter, batteries) have decreased. The price reduction is, however, less for off-grid equipment that is available from local retailers than it is for these components in general. In our work, price reductions affect the total system cost to a less extent than they affect new systems, as many components were already available. We explore the effects of the decreasing prices in our sensitivity analysis by varying the battery price.

Performance losses related to the PV panels such as temperature effect, mismatch, degradation, wiring and soiling were combined into one derating factor in the model. The value was derived in [48] from measured and simulated PV output values, and hence represents the average efficiency during the measurement period. The factor was further adjusted to compensate for losses in the charge controller and further degradation over the project lifetime. The generator was modelled with automatic start. The minimum load ratio was selected based on specifications of the real generator and affects the generator's minimum output power and its simulated operation pattern.

Battery efficiency was estimated through measurements. The maximum depth of discharge (DOD) was defined as being consistent with the low-voltage disconnection settings of the charge controller and the inverter. A typical battery throughput and cycle life for the battery type was applied. The battery float life was shortened by two years in relation to battery manufacturer specifications as a way of compensating for higher ambient temperature than under standard conditions. The purchase of new batteries was assumed at the start of simulations as the current battery bank is approaching the end of its lifetime. For system

Table 1
Specifications of the components in the PV-diesel hybrid system that has been used as a model for simulations.

System components	Capacity	Year of installation	Additional information
PV	655 W_P	2008 (some older)	36 PV cells connected in parallel, Tilt: 23°, Orientation: North
Generator	12 kW	2008	Manual start only
Batteries	1000 Ah 12 V	2011	Five 200 Ah, 12 V batteries in parallel Type: AGM, sealed deep-cycle lead-acid
Bi-directional inverter	Inverter: 1200 W Rectifier: 840 W	2015	Xantrex DR 230 VAC/50 Hz Mid-range Inverter/Charger
Charge controller	60 A	2011	Outback MX 60, 60 A
Load	DC: 0.34 kW h/d AC: 1.12 kW h/d	From 2008 onwards	DC: Lights and security lights all night AC: Computers, printers, phone charging, TV, lights

Table 2
Input parameters for modelling.

Category	Item	Technical parameters	Economic parameters
Components	PV panels	Derating factor: 0.59	Investment/Replacement cost: -
	Generator 12 kW	Minimum load ratio: 30%	Investment/Replacement cost: -
	Generator 1.2 kW[a]	Minimum load ratio: 90%	Investment/Replacement cost: -
	Batteries	Battery efficiency: 80% Maximum depth of discharge: 60% Float life: 10 years Lifetime throughput: 466 kW h Cycle life: at 30% DOD – 700 cycles, at 50% DOD – 400 cycles, at 100% DOD – 150 cycles	Investment/Replacement cost: 1750 USD
	Power converter	Component lifetime: 10 years Inverter efficiency: 85% Rectifier efficiency: 80%	Investment cost: - Replacement cost: 1680 USD
	Rectifier[a]	Component lifetime: 10 years Efficiency: 80%	Investment cost: - Replacement cost: 120 USD
	Replacement of DC appliances to AC[a]		Investment cost: 50 USD
Grid	Grid connection		Connection fee: 520 USD
	Purchase of electricity		Stand by fee: 2.75 USD/month Electricity charge: 0.151 USD/kW h
	Sale of electricity		Tariff, Dec-Jul (wet): 0.081 USD/kW h Tariff, Aug-Nov (dry): 0.108 USD/kW h
Energy sources	Solar irradiation	Yearly average: 5.58 kW h/m^2, day	
	Diesel price		0.83 USD/l
System parameters	System lifetime	20 years	
	Interest rate		6% annual

[a] Components that were not originally a part of the system but that have been used in some simulations. The parameters are further explained in the respective sections of 2.5 Varying boundary conditions and 2.6 System configurations.

configurations without batteries, HOMER's operation strategy *load following* was used. For systems with batteries, *cycle charging* was applied with a set-point state of charge of 80%. Further explanations of the operation strategies can be found in the user manual available from HOMER [33].

The PVDMG has two units for power control and conversion: a bi-directional inverter and a solar charge controller. In HOMER, all power converters are modelled as one unit. This limits the possibility to study lifetime, efficiency and price variations for charge controllers and inverters in great detail. The power converter was simulated according to specifications for the bi-directional inverter given in Table 1. In the model, the inverter and rectifying efficiencies are always constant. The values were determined through monitoring, and thus represent the average efficiency of the bi-directional inverter during the measurement period. Losses in the charge controller were included in the derating factor for the PV panels. With the real inverter, the grid and generator cannot be used at the same time. In the model, this is possible. Since the generator is not used when grid power is available, this has no impact on the validity of the simulated system operation.

2.4.2. Grid

Fees associated with grid connection and tariffs for purchase and sales of power from/to the grid are specified in Table 2. Prices are taken from the Tanzanian Electric Supply Company (TANESCO) and the Tanzanian Energy and Water Utilities Regulatory Authority (EWURA) [49,50]. The Small Power Producers Framework includes standardized power purchase agreements for renewable energy systems with a capacity of 100 kW to 10 MW, intended to supply commercial electricity to the national grid [11,51,52]. Although they do not apply to the studied PVDMG, the tariffs have been used for the purpose of investigating the impact on system economics of selling excess power to the grid. Incentives that apply to grid connection and the purchase of electricity have been included in the model by using the subsidized prices charged to the end users. Our price model is thus time-bound and geographically specific.

The used simulation software does not have any inbuilt function for direct modelling of blackouts in the national grid. Blackouts have therefore been modelled indirectly by applying an extremely high rate for power purchase combined with no sales, which makes HOMER disregard grid electricity purchase. The resolution for different rates in HOMER is limited to full hours, and can be specified for either weekdays, days at the weekend or all days within a month. The approach hence allows for specification of deterministic blackouts, but does not account for random interruptions in the grid. Variations in grid availability has been studied by application of different blackout scenarios described in Section 2.5.2. Fig. 3 shows an example of simulated blackout periods.

2.4.3. Energy sources

Solar irradiation data for the specific location provided by NASA [53] (satellite data) was used in the model. The data has been validated in an earlier publication [48] by comparison with measurements at the location as well as satellite data and ground-based data for the nearby city Mwanza provided by Meteonorm [54]. The diesel price was taken from a local fuel distributor.

2.4.4. Load profiles

Load profiles have been derived from monitored data from 2011 to 2013. The average AC load was 1.1 kW h/day, with monthly averages ranging from 0.6 to 1.6 kW h/day. The average DC load was 0.3 kW h/day, with variations from 0.2 to 0.6 kW h/day. There were no indications of long-term changes or seasonal differences either in AC or DC load that had to be considered. To create load profiles, months with a typical average load were selected. The AC load profile of the chosen month was adjusted slightly to better conform to typical differences between weekends and weekdays. The daily load profiles used in the model are shown in Fig. 4 (AC) and Fig. 5 (DC). Their values are 1.09kW h/day for AC and 0.35 kW h/day for DC, resulting in a total energy consumption of 522 kW h/year. Monitored daily and monthly variations in load were represented in the model through the definition of two parameters in HOMER: *random variability from day-to-day* and *random variability from time-step-to-time-step* (for further explanations,

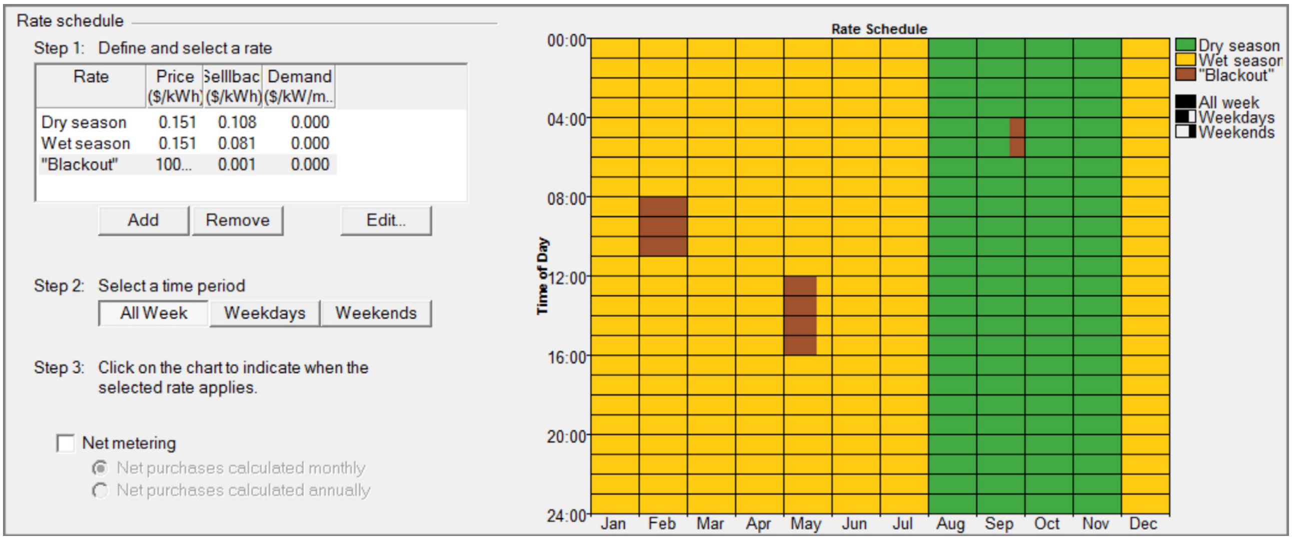

Fig. 3. An example of a rate schedule used in HOMER to simulate blackouts. The brown fields indicate when blackouts occur: every day in February from 08:00–11:00, from Monday to Friday in May from 12:00–16:00, and on Saturdays and Sundays in September from 04:00–06:00 [45]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

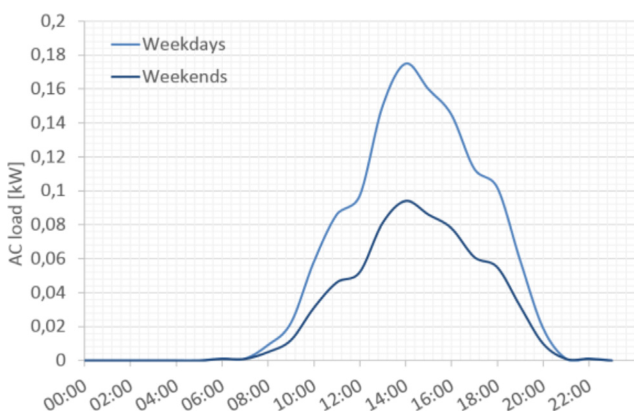

Fig. 4. The daily AC load profiles used for simulations. Average AC load: 1.09 kW h/day.

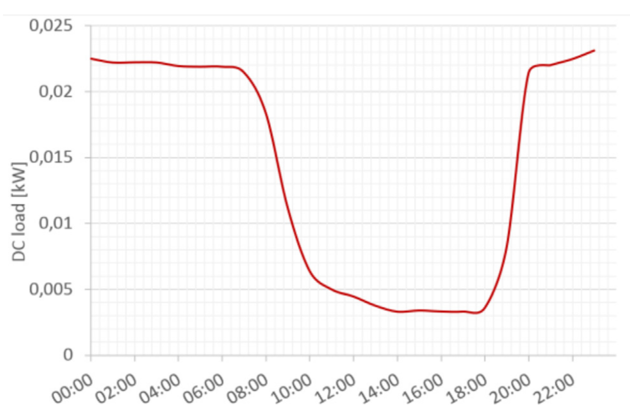

Fig. 5. The daily DC load profile used for simulations. Average DC load: 0.34 kW h/day.

see [33]). For AC, the variabilities used in simulations were 48% and 66% respectively. For DC, 33% was used for day-to-day variability and 29% for time-step-to-time-step variation. Further details on load distribution can be found in [55].

2.5. Varying boundary conditions

This section describes the variations in boundary conditions applied to the PVDMG in simulations.

2.5.1. Load

Load data for the PVDMG and the modelling of the load profiles are described in Section 2.4. The electricity usage depends strongly on behaviour patterns and can look very different if new activities are added or others disappear. Limitations in accommodating changes in load is a general challenge associated with small power systems. In order to investigate how a load increase would affect the results, two additional load levels have been used in the simulations. The actual load (522kW h/year) is denoted as L. L2 and L4 stand for twice and four times the actual load respectively. L2 and L4 have the same relative load distribution over time as L. These three load levels applied to the current PVDMG will act as base cases when studying blackouts and system modifications.

2.5.2. Grid availability

Two major parameters were considered in the modelling of blackouts: duration and time-distribution of blackouts. The effect of duration was studied for blackouts occurring in consecutive periods from 0 to 24 h per day, evenly distributed over day and night over the months. For 0.5 and 0.25 h of blackouts per day, full hours every day in every other/every four months were simulated. The study on time distribution was done by modelling the four blackout scenarios in Table 3. The total number of blackout hours for the first three scenarios are somewhat lower than the average in Tanzania (648 h in 2013) [14]. However, grid reliability has improved by 3–4% per year over recent years, and the used values are hence realistic for the near future. The fourth scenario was modelled to correspond to an extremely unreliable grid, as grid reliability varies drastically depending on location in Tanzania.

2.5.3. Generator size

The generator was, as mentioned, initially sized to supply workshop machines. If operated to supply the hybrid system, it is substantially oversized. The oversizing results in poor fuel efficiency and unreasonable fuel costs. Simulations have thus also been conducted with a smaller 1.2 kW generator in order to evaluate the impact of the generator size on the cost savings for different system configurations and with different boundary conditions. In order to imitate realistic manual

Table 3
Blackout distribution scenarios.

Scenario	Time (h/year)	Description
Blackouts at weekends	408	Blackouts on all Saturdays and Sundays consecutively for two months (April and September) per year
Blackouts on weekdays	408	Blackouts on all Mondays to Fridays consecutively for one month (April) per year
Blackouts for one hour/day	365	Blackouts for one hour per day throughout the year, distributed evenly over the day's 24 h
Blackouts for four hours/day	1460	Blackouts for four consecutive hours per day throughout the year, distributed evenly over the day's 24 h

Table 4
The considered system configurations.

System variant	Description
Grid only	Abandonment of the PVDMG and connection to and reliance on the national grid alone. The local power distribution network remains unchanged
Grid only, AC only	Same as "Grid only", but with the local DC distribution system abandoned and the local AC power distribution network increased to provide energy services previously provided by DC
Grid-connected, With batteries	Connection of the PVDMG to the national grid, thereby forming a PV-diesel-battery-grid system
Grid-connected, No batteries	Connection of the PVDMG to the national grid without using batteries, thus forming a PV-diesel-grid

operation, the minimum load ratio of the 1.2 kW generator was set to 90%, resulting in fewer starts and stops of the generator than if a lower ratio were to have been used. No investment cost applies to the 1.2 kW generator in order to obtain results comparable to those from simulations with the 12 kW generator.

2.6. System configurations

The intention of this study was to investigate which parts of the PVDMG might be reasonable to keep in operation when grid-connection becomes available so that a redesign of the system can be avoided. The different system configurations used for simulations are described in Table 4.

The two system configurations relying on the national grid alone (grid only; grid only, AC only) contain some other components than does the original PVDMG. In grid only, the bi-directional inverter was replaced by a rectifier at the time of replacement (*Rectifier* in Table 2). In the system configuration grid only, AC only does not have a rectifier. Initial costs were added for AC appliances to replace DC appliances (*Replacement of DC appliances to AC* in Table 2). Revenues from selling no longer used equipment was not included. In some simulations, the sale of power from the PVDMG to the grid has been considered.

3. Results

Results from simulations with the system modifications and variations in boundary conditions described in the previous chapter are here presented as answers to the following three questions.

What is the economic viability of the evaluated system configurations if

1. continuous power supply from the national grid is assumed?
2. blackouts in the national power grid with variations in time and distribution are considered?
3. if a more appropriately sized generator was used?

3.1. Continuous grid supply (no blackouts)

Here, all system configurations in Table 4 and all three load levels L, L2, L4 have been used. For the two system configurations using the PVDMG, the sale of excess power has been considered as separate cases. Continuous power supply from the national grid was assumed. Fig. 6 shows the savings in COE for the simulated system configurations, relative to the base case at the respective load level.

As shown in Fig. 6, the most economically viable option for all load levels is either to use the national grid only or to connect the PVDMG without batteries to the national grid. With continuous power supply from the national grid, the batteries were associated with costs, but without any benefits in reduction of generator use or improved energy access. The influence of the batteries on the COE decreases with increasing load, although the battery cost in absolute terms is equal for all load levels. Replacement of batteries was required at the end of the component lifetime for all simulated load levels.

The two system configurations with grid only and the grid-connected configurations without batteries have very similar savings. One could expect the grid-connected configurations to have a higher saving as the existing (paid-off) system reduces the amount of purchased energy significantly. Nonetheless, these savings are compensated by replacement costs of the charge controller and inverter (1680 USD).

When only the national grid is used, there are no benefits to maintain the DC distribution grid. Instead, converting the DC distribution to AC would eliminate conversion losses and the need for a rectifier, and the same type of AC light bulbs could conveniently be used for the whole system.

A factor that in Fig. 6 is shown to affect the system economy only marginally is whether excess energy is sold or not. At the load level L, the amount of sold energy is equal to as much as 60% of the load. Still, the economic effects are minor. With increasing load, the saleable power decreases. An operational strategy that better utilizes the batteries could decrease the electricity purchased from the grid, but might instead increase the need for battery replacements due to more frequent cycling.

3.2. Blackouts with variations in time and distribution

3.2.1. Daily blackouts for 0–24 h

For all three load levels L, L2 and L4, simulations have been carried out with daily blackouts occurring in consecutive periods from 0 to 24 h per day. Grid-connected PVDMG with and without batteries has been compared to the base case. Fig. 7 shows the results at load level L4. Using the grid only was not considered due to its inability to provide power during blackouts, but the cost when no blackouts occur is shown as a reference point. In Fig. 8, coloured fields indicate which system configuration is the most economically beneficial at each combination of hours of blackouts and load.

Figs. 7 and 8 show that for low duration of blackouts (around 0.5 h per day, varying slightly with load levels), the most economically viable system configuration is grid-connected PVDMG without batteries (green area in Fig. 8). As can be seen in Fig. 7, the saving in COE for the grid-connected PVDMG without batteries decreases dramatically with the number of blackout hours. The cost decrease is even more predominant at lower load levels, and is due to inefficient generator

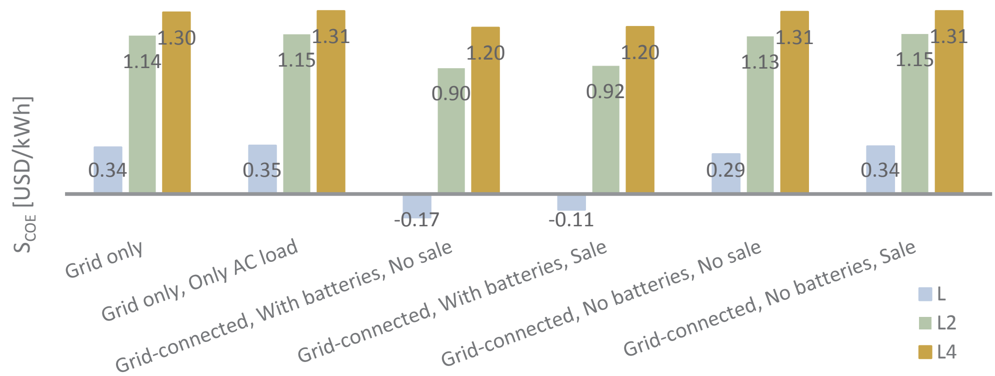

Fig. 6. Savings in COE relative to respective base case for different system solutions and different load levels.

operation. For loads lower than around L1.1 (slightly above the actual load level L), base case operation (red area in Fig. 8) is more economically beneficial than grid-connection of the PVDMG with batteries, regardless of blackouts. For loads above L1.1 and blackouts above 0.75 h per day, the grid connected PVDMG with batteries is the most feasible solution (blue area).

3.2.2. Distribution of blackouts

The blackout scenarios in Table 3 have been used to further study the effect of the distribution of blackout hours. The grid-connected PVDMG was modelled with and without batteries and load level L2. System configurations relying on the national grid only were excluded due to the inability to provide continuous power. The results are shown in Fig. 9.

As Fig. 9 shows, the savings in COE relative to the base case were similar for all used blackout scenarios when including batteries, regardless of how the blackouts were distributed. The main reason for the economic savings is the longer battery lifetime obtained when power from the national grid is used. The batteries did not reach the maximum number of cycles before the end of float life in any of the simulations. For most cases, the slight cost differences were due to different amounts of energy being bought from the grid. Only when blackouts occurred on full weekdays for one month could the batteries not sustain the loads during all blackout hours and the generator was operated.

For the system configurations without batteries, more frequent use of the generator to supply power during blackouts leads to a clear decrease in savings. For the four-hour blackouts per day, the COE is even higher than in the base case. The difference in savings for the two blackout scenarios including the same number of blackout hours is due to a lower load during weekends, thus lower generator usage.

3.3. Generator size

In this section, the consequences of using a smaller generator of 1.2 kW instead of 12 kW have been studied. The PVDMG without grid connection has been simulated with both generators for all three load levels. The results are presented in Table 5. Results from simulations at load level L2 with the two generators, system modifications and various blackout scenarios are presented in Fig. 10.

The results in Table 5 show clearly that the 12 kW generator is oversized. This leads to poor fuel efficiency and a vast amount of excess electricity, which in turn has significant effects on the COE. The simulation results with a 1.2 kW generator show clear performance improvements for all three load levels while still fulfilling the load (energy) demands.

Fig. 10 shows the savings in COE for both generators relative to the base case with the 12 kW generator at load level L2. For system configurations resulting in no or low generator utilization, i.e. where batteries provide sufficient backup in case of a blackout, the generator type does naturally not affect the COE that much. The small differences visible in Fig. 10 are due to operational differences in the simulation. When no batteries are used, the generators operate every time a blackout occurs and the load cannot be supplied with PV only. Consequently, the economic benefits of using the smaller generator increase with the increasing number of blackout hours.

At load level L2 and with the 12 kW generator, it is more economically viable to use the off-grid PVDMG than to connect it to the grid without using batteries, if blackouts occur for more than 2.75 h per day. The same breaking point for the 1.2 kW generator occurs at slightly more than 5 h per day (Fig. 10 and further analysis). At this load level (L2), off-grid operation of the PVDMG is not the most cost-effective with any of the two generators though. The simulations show that to

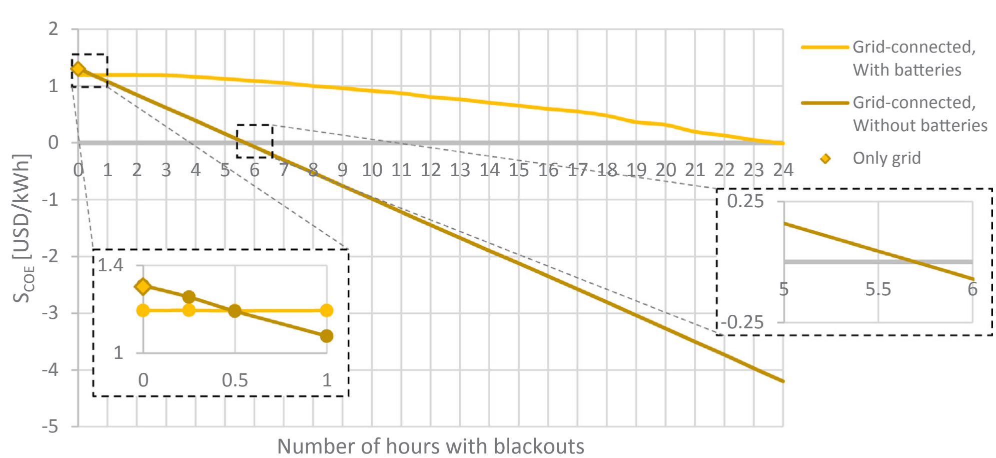

Fig. 7. Savings versus number of hours with blackouts for grid-connection with and without batteries. Load level L4.

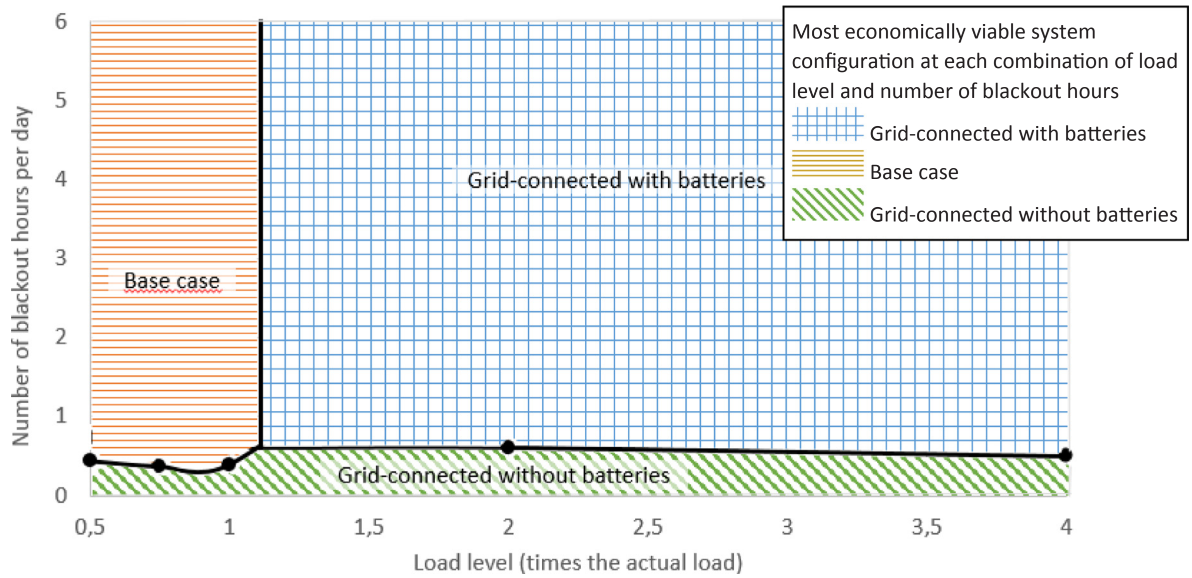

Fig. 8. Coloured fields indicating which system configuration is the most economically beneficial with different numbers of blackout hours and different load levels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

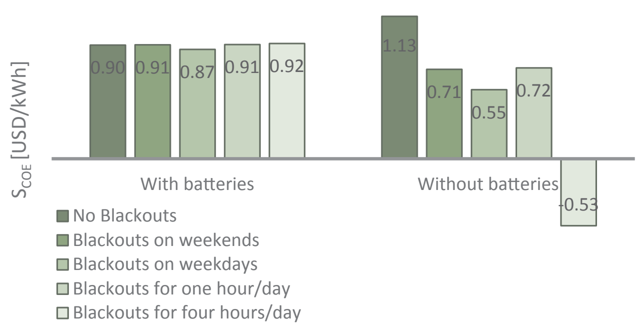

Fig. 9. Savings relative to base case for the PVDMG (with and without batteries) connected to the national grid with different blackout scenarios. Load level L2. Details about the different blackout scenarios are found in Table 3.

Table 5
Simulation results for the PVDMG for three load levels and two generator sizes.

Load level	L		L2		L4	
Generator size [kW]	12	1.2	12	1.2	12	1.2
Load [kW h/yr]	522	522	1044	1044	2088	2088
Electrical production generator [kW h/yr]	43.2	19	1894	588	4788	1825
Fuel efficiency [kW h/l]	1.9	3.1	1.9	3.0	1.9	3.0
Hours of operation [h/yr]	12	16	526	499	1330	1556
Generator starts [starts/year]	2	8	91	256	279	603
Electrical production PV [kW h/yr]	759	759	759	759	759	759
Excess energy [kW h/yr]	161	136	1241	28.5	2743	114
Unmet load [kW h/yr]	0	0	0	0	0	0.3
COE rel. to 12 kW generator [USD/kW h]	–	−0.03	–	−0.72	–	−0.91

connect the PVDMG to the grid either with or without batteries is more beneficial for all blackout scenarios. Fig. 10 combined with further analysis shows that the number of daily blackout hours at which it becomes economically beneficial to include batteries increases from 30 min with the 12 kW generator to 3 h with the 1.2 kW generator.

3.4. Comparison of results

Here, selected results from previous sections are combined so that their internal relations can be studied further. Fig. 11 shows the COE for a set of different system configurations and blackout scenarios. For visualization purposes, all costs are shown in relation to the COE for the base case at load level L1, which has been normalized to one.

As shown in Fig. 11, the COE for the base case increases with increasing load. The major reasons for this are increased generator operation and more frequent battery replacements. For the grid-connected system configurations, however, the COE decreases with increasing load. The grid connection fees and the non-changing component costs become distributed over more kW h. In configurations with batteries, the generator size does not affect the COE much since the load can almost always be covered without the use of the generator. For system configurations without batteries, the generator size has a much greater impact on the COE as it has to be operated to supply the load during blackout hours. Here, low load can – during some blackouts – be completely covered by PV. For most blackout occasions, however, the generator has to be operated to cover the load. The lower the load, the greater the cost will be for each kW h of load to operate the generator.

3.5. Summary of results

For the studied case, there are few benefits to maintain the existing PVDMG for a grid without blackouts. The best configuration of the PVDMG without batteries will result in the same savings as if only the grid were being used and the PVDMG were abandoned. Decisive for this result were the low grid connection fees caused by the PVDMG's proximity to the national grid and subsidies to grid connection and power purchase that applied in Tanzania, and the relative low revenues for feeding electricity to the grid. However, as equipment costs are decreasing (PV, inverter, batteries) and as more market-based electricity tariffs can be expected in the future, a grid-connected PVDMG with batteries may become beneficial.

Relying on the national grid only is only an alternative if no blackouts occur, or if interrupted electricity access can be accepted. As expected, when blackouts exist, it is beneficial to maintain the PVDGM. The duration of the blackouts and the load level determine which system configuration is most feasible. For small load levels, it is most beneficial to maintain the system as it is without connecting it to the

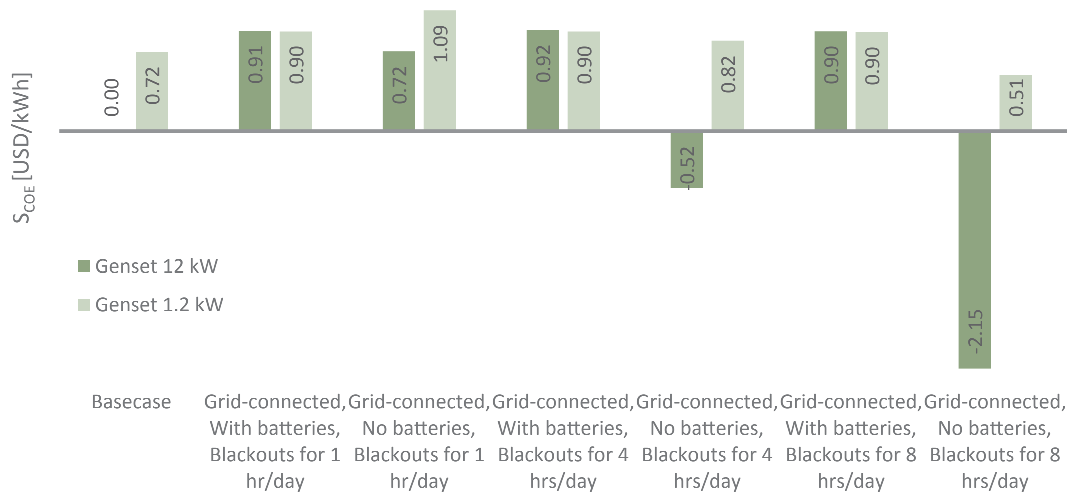

Fig. 10. Savings relative to the base case with the 12 kW generator for different blackout scenarios and different system configurations. The load in the simulations is L2.

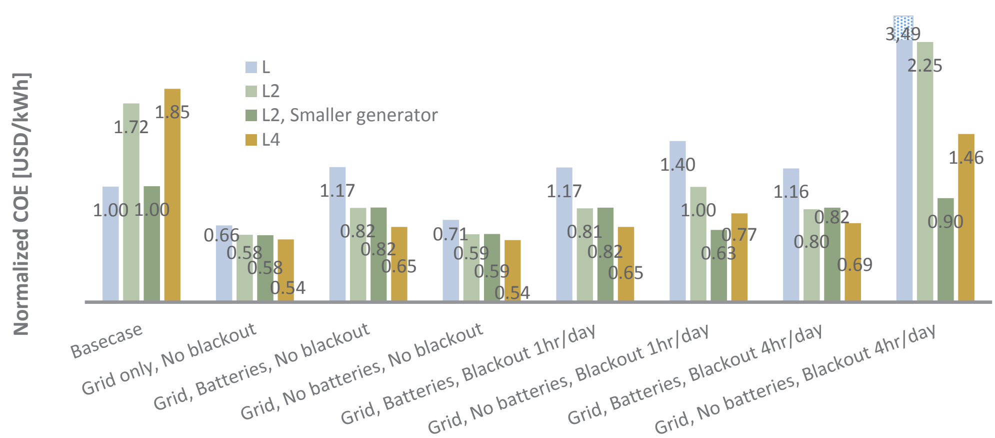

Fig. 11. COE normalized to COE for the base case with the actual load. Sales of excess energy not included.

grid. For higher load levels and relatively short blackouts (less than one hour), the PVDGM should be grid-connected but without batteries. The grid-connected PVDGM with batteries is the best configuration for blackouts of more than 1 h at a load level higher than the current load. The distribution of the blackouts has very little impact on the system configuration. Using a more appropriately sized diesel generator would improve the feasibility of the grid-connected PVDGM without batteries for blackouts longer than one hour.

4. Discussion

4.1. Sensitivity of results

With continuous grid supply, the two system configurations “grid only” and “grid-connected without batteries” resulted in almost identical savings in COE in relation to continuous base case operation. Analysis of the effect of variations in the grid tariff shows that for load level L2, an increase in tariff by 50% would increase the COE of the grid only solution by 32% while it would increase the COE for the grid-connected PVDMG without batteries by only 20%. The savings would thereby be slightly higher for the latter system (1.107 USD/kW h compared to 1.077 USD/kW h for “grid only”). Similar dependencies were seen for the other two load levels.

With continuous grid supply, the batteries have no function in the grid-connected PVDMG. A lower battery cost will decrease the COE, but the savings will never reach those achieved if batteries were not used at all. In a situation with blackouts in the national grid, batteries and generators compete for being the most cost-competitive alternative to provide backup power during blackouts. As long as the (oversized) 12 kW generator is being used, it is beneficial to keep the batteries as a backup source for almost all simulated blackout scenarios and load levels. Decreasing battery prices will support this further. However, if a (more appropriate) 1.2 kW generator is used, the generation losses will decrease and improve the feasibility of the generator as a backup source (as Fig. 10 shows). For example, for blackouts of 1 h/day and a load level L2, the PVDMG without batteries and using a 1.2 kW generator will give the highest savings. If battery prices decrease by 50%, the saving for the same PVDMG with batteries would only increase by 13% but still have 0.07 USD lower savings compared to the PVDMG without batteries and the 1.2 kW generator. Higher diesel costs (100%) would decrease the savings of the PVDGM without batteries by 0.04 USD if the 1.2 kW generator is used and by 0.42 USD with the 12 kW generator. Therefore, for shorter blackouts, it is more beneficial to use only a 1.2 kW generator as a backup source instead of batteries independent of expected price variations. For longer blackouts and higher load levels, batteries are always beneficial as backup sources combined with the generator independent of which of the two generators that is chosen and independent of the expected price developments.

4.2. Limitations and simplifications

A consequence of using a real and existing system is that our results are impacted by peculiarities associated with the specific system. The study gives a rather realistic picture of aspects to consider if retrofitting an existing PVDMG to conform with a national grid. By varying load and generator size and through a sensitivity analysis, we have been able to address the effects of some case-specific parameters. Despite the case-specific nature of figures and numbers presented as results, the overall trends and aspects pointed out as being important to keep in mind are hence highly relevant.

Our work carries some limitations related to assumptions in our model, to our evaluation method and to the used simulation tool. Here, we further reflect upon some of these issues.

– HOMER performs simulations for one year, and uses the results to calculate lifetime costs and performances. Consequently, component degradation over time is disregarded. Replacement intervals are defined manually or by specification of energy throughput (for batteries). PV and power conversion efficiencies are defined as constant, thus load dependencies are neglected. Our efficiency values were derived from measurements at load level L. We have not been able to verify through measurements the accuracy of these values for the other simulated load levels.
– In the simulations, the generator starts automatically, while in reality it must be started manually. The simulations hence give a picture of how the generator should be operated to achieve optimum system performance, rather than representing the actual operation. The evaluation does not show or value the power interruptions that would occur at times of blackouts in the grid if using a manually started generator as the only backup source.
– Savings in COE compared to a base case (off-grid PVDMG) were used as the main evaluation method for the feasibility of different system configurations. However, this method does not reflect on environmental aspects such as carbon emissions. The method also values every kW h equally while one could give a higher value for each kW h of load supplied during blackouts. Rating these kW h higher would certainly make the PVDMG more valuable.
– Factors related to the financing of the system have been considered to a limited extent. The discount rate is used to represent the financing mechanism, but this has not been investigated in great detail. Other possible incentives than those inbuilt in the cost figures for grid connection and purchase of electricity from the grid – for example, subsidies and grants obtainable from institutions working to promote rural and renewable electrification – have not been considered.

4.3. Results in comparison with previous studies

Previous studies [37,38,40] conclude that it is not economically feasible to install PV or PV hybrid solutions for backup purposes. In contrast to our study, these studies include investment costs for the PV systems. In our case, the PVDMG was already installed and in operation. Accordingly, only operational expenses and component replacements were considered. Under these circumstances, the utilization of a PV or PV hybrid power system for backup or complementary purposes is interesting.

In accordance with our argumentation, Twaha et al [39] and Murphy et al [40] conclude that future price changes speak in favour of PV and PV hybrid systems being used in conjunction with national power grids. They expect an increase in fuel costs and a decrease in PV system component prices. In our study, increasing fuel prices and increasing fees for grid connection and power tariffs in the national grid point in the same direction.

In our work, blackouts were modelled using a price model that avoided the use of the grid. Murphy et al [40] simulated blackouts in HOMER by replacing the grid with a generator carrying the characteristics of the grid. The two methods result in equal possibilities to model blackouts and entail the same limitations in how random variations or, for example, long-lasting blackouts can be modelled. Murphy et al [40] derive blackout patterns based on statistics. This improves the reflection of the real grid availability, at the same time as the possibility to systematically investigate blackout scenarios is reduced.

In general terms, we agree with Cader [35] and other researchers who highlight the importance of considering blackouts in the national grid in the process of investigating different electrification strategies. In line with our results, Rajbongshi et al [37] and Mahapatra and Dasapa [38] find that the preferable choice of electrification strategy among different system configurations depends on grid availability among other factors.

5. Conclusions

A PVDMG in Tanzania was studied in order to evaluate if and under what circumstances it is beneficial to connect the PVDMG to the national grid. The results show that the answer is strongly dependent on the load level and the duration of blackouts in the national grid. Without blackouts, there are few benefits to maintain the PVDMG despite the fact that the existing system does not carry any investment costs. When blackouts occur, the extent of blackouts and the load on the system determine which system configuration is most feasible among the following alternatives that were studied: continuous operation of the off-grid PVDMG, or grid connection of the PVDMG with or without batteries. We also show that sales of excess electricity from the PVDMG affect the COE to a limited extent with the current prices and excess power, and that variations in battery and diesel prices affect the choice of system configuration to a lesser extent than the load and the extent of blackouts.

Despite the widely promoted two-track electrification approach, very little attention has been given to the interplay between existing decentralized power systems and expanding national power grids. Apart from being beneficial for the system owners and users, as we show in our case study, grid-connecting decentralized systems can help to cover the overall increasing electricity demand by promoting self-consumption and sales of excess electricity from decentralized systems. This topic deserves attention at a local level from researchers, project implementers and system owners in order for existing systems to be used wisely when a national grid becomes available. At a policy level, the interaction needs attention so that there can be assurance that regulations, incentives and support both facilitate overall electrification goals and encourage promoted strategies.

With the exception of studies investigating suitable backup solutions, blackouts in the national grid are often disregarded in studies and evaluations of different possible rural electrification options. Our results show clearly that depending on the extent to which blackouts occur, different system configurations are feasible. Figures that result from investigations related to one specific system may not be valid for another; however, those working with proposals or decisions regarding electrification can still use our results as a basis for investigations in their specific cases.

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